

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

Master of Business Administration (MBA)

Post Graduate Diploma in Management (PGDM)

Semester – IV University Examination, May - 2018

MB806.3: Logistics and Supply Chain Management

Date: 16.05.2018, Wednesday

Time: 10:00 A.M. to 01:00 P.M.

Maximum Marks: 70

Instructions:

1. This paper contains six questions, of which the last one is a case analysis.
2. All questions are compulsory.
3. The first five questions will carry 10 marks each and the last one (case analysis) will carry 20 marks.

- Q- 1** “Supply chain management encompasses critical decisions that affect the profitability of an organisation.” Explain the given statement in the light of various decisions related to supply chain of an organisation which can have tremendous impact on its profitability. **10**
- Q-2 (a)** Define the following terms with suitable example: **05 10**
- i. Mean Absolute Deviation (MAD)
 - ii. Chase Strategy
 - iii. Standard Deviation of Demand
 - iv. Pipeline Inventory
 - v. Cost of Under-stocking
- Q-2 (b)** “Inventory Management under uncertain demand and lead time is very much different than EOQ model.” Differentiate the Stochastic and Generalist models of inventory management in the light of re-order point calculation, safety stock assumptions, and decisions regarding ordering quantity. **05**
- Q- 3** “Manufacturers have wide variety of transportation modes available to transport their products.” Explain various options available with manufacturer and their effect on key inventory parameters along with advantage and dis-advantage. **10**
- Q-4** A manufacturing company has the following actual demand for last three years: **10**

Time Period	Quarter	Actual Demand
1	1	398
2	2	395
3	3	361
4	4	400
5	1	410
6	2	402
7	3	378
8	4	440
9	1	465
10	2	460
11	3	430
12	4	473

What forecasting model can be used to forecast the demand for the data given above? Justify your answer. Forecast the demand for next four time period (Time Period 13, 14, 15, & 16).

Q- 5 “Revenue management plays a very crucial role in determining the profitability of a firm.” Define revenue management and answer the following questions:

10

- i. Why does fashion industry offer price discounts close to the end of the season?
- ii. Some researchers argue that application of revenue management resulted in increase in capacity utilisation but resulted in reduction in total profits. What could be the reason for the same?
- iii. Explain why dynamic pricing provides higher profits compared to fixed price strategy?

Q-6

Case # 1: Brushing Up On Inventory Control

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Robert works as an inventory control manager at Nightingale Drugstore. One fine evening, while watching television, he was exposed to advertisement of Totalee toothbrushes.

The commercial concludes that the customer should buy a Totalee toothbrush because the toothbrush is Totalee revolutionary and Totalee effective. It certainly is effective; it is the most popular toothbrush on the market! At that moment, with the inventory article and the toothbrush commercial fresh in his mind, Robert experiences a flash of brilliance.

He knows how to control the inventory of Totalee toothbrushes at Nightingale Drugstore! As the inventory control manager at Nightingale Drugstore, Robert has been experiencing problems keeping Totalee toothbrushes in stock. He has discovered that customers are very loyal to the Totalee brand name since Totalee holds a patent on the toothbrush endorsed by 9 out of 10 dentists. Customers are willing to wait for the toothbrushes to arrive at Nightingale Drugstore since the drugstore sells the toothbrushes for 20 percent less than other local stores. This demand for the toothbrushes at Nightingale means that the drugstore is often out of Totalee toothbrushes.

The store is able to receive a shipment of toothbrushes several hours after an order is placed to the Totalee regional warehouse because the warehouse is only 20 miles away from the store. Nevertheless, the current inventory situation causes problems because numerous emergency orders cost the store unnecessary time and paperwork and because customers become disgruntled when they must return to the store later in the day. Robert now knows a way to prevent the inventory problems through scientific inventory management!

Because rush hour traffic has dissipated, the drive to the drugstore takes Robert no time at all. He unlocks the darkened store and heads directly to his office where he rummages through file cabinets to find demand and cost data for Totalee toothbrushes over the past year.

Aha! Just as he suspected! The demand data for the toothbrushes is almost constant across the months. Whether in winter or summer, customers have teeth to brush, and they need toothbrushes. Since a toothbrush will wear out after a few months of use, customers will always return to buy another toothbrush. The demand data shows that Nightingale Drugstore customers purchase an average of 250 Totalee toothbrushes per month (30 days).

After examining the demand data, Robert investigates the cost data. Because Nightingale Drugstore is such a good customer, Totalee charges its lowest wholesale price of only \$1.25 per toothbrush. Robert spends about 20 minutes to place each order with Totalee. His salary and benefits add up to \$18.75 per hour. The annual holding cost for the inventory is 12 percent of the capital tied up in the inventory of Totalee toothbrushes.

Answer the following questions:

- i. Robert decides to create an inventory policy that normally fulfills all demand since he believes that stock-outs are just not worth the hassle of calming customers or the risk of losing future business. He therefore does not allow any planned shortages. Since Nightingale Drugstore receives an order several hours after it is placed, Robert makes the simplifying assumption that delivery is instantaneous. What is the optimal inventory policy under these conditions? How many Totalee toothbrushes should Robert order each time and how frequently? What is the total inventory cost per year with this policy? 07
- ii. Totalee has been experiencing financial problems because the company has lost money trying to branch into producing other personal hygiene products, such as hairbrushes and dental floss. The company has therefore decided to close the warehouse located 20 miles from Nightingale Drugstore. The drugstore must now place orders with a warehouse located 350 miles away and must wait 6 days after it places an order to receive the shipment. Given this new lead time, how many Totalee toothbrushes should Robert order each time, and when should he order? 03

Case # 2: Inventory Management with Quantity Discounts

MBI is a manufacturer of personal computers. All its personal computers use a 3.5-inch high-density floppy disk drive which it purchases from Ynos. MBI operates its factory 52 weeks per year, which requires assembling 100 of these floppy disk drives into computers per week. MBI's annual holding cost rate is 20 percent of the value (based on purchase cost) of the inventory. Regardless of order size, the administrative cost of placing an order with Ynos has been estimated to be \$50. A quantity discount is offered by Ynos for large orders as shown below, where the price for each category applies to every disk drive purchased.

Discount Category	Quantity Purchased	Price (\$ Per Disk Drive)
1	1 to 99	\$100
2	100 to 499	\$95
3	500 or more	\$90

Answer the following question:

- i. Determine the optimal order quantity according to the EOQ model with quantity discounts. What is the resulting total cost per year? With this order quantity, how many orders need to be placed per year? 10